

TECHNICAL ASSESSMENT REPORT

FREIGHT RATE PREDICTION

Machine Learning Engineer Assessment Report



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Organization	Spotter
Submission Date	31 August 2026
GitHub Repository	github.com/equadri17/freight-rate-prediction
Final Selected Model	CatBoostRegressor with <code>log1p(posted_rate)</code>

Executive Summary

Problem, key findings, model selection and final recommendation

Problem: This assessment develops a supervised machine learning regression model to predict freight posted rates for unseen transportation loads. The labeled development dataset contains 48,000 loads, while the final validation dataset contains 12,000 unseen loads. A separate 31-row December 2025 scenario is also evaluated to demonstrate the model's response to date changes under fixed load conditions.

Key Findings: Exploratory analysis identified transportation distance as the strongest simple numerical predictor of posted rate, with a Pearson correlation of approximately 0.9085. The target distribution is strongly right-skewed, while equipment type, route/location and time-related variables also provide meaningful predictive information.

Model Selection: Five approaches were benchmarked: a median-rate baseline, Ridge Regression, Random Forest, CatBoost with the raw target, and CatBoost with a log-transformed target. The log-target CatBoost model substantially outperformed the earlier approaches. Targeted hyperparameter tuning then selected a configuration using 300 iterations, depth 6, learning rate 0.05 and L2 regularization 5.

Final Result: On the chronological October 2025 holdout, the selected model achieved an MAE of \$107.70, RMSE of \$648.20 and R² of 0.8202. The model was subsequently retrained on all 48,000 labeled development observations and used to generate predictions for all 12,000 unseen validation loads.

Recommendation: The recommended model for the assessment submission is the tuned CatBoostRegressor with a log1p-transformed target. The approach provides the lowest validation MAE among the tested configurations while retaining a clear, reproducible feature-engineering and validation pipeline. The hidden final evaluation score should be treated separately from the development holdout metrics and will be determined by the assessment evaluator after submission.

Executive Performance Snapshot

Holdout MAE	Holdout RMSE	Holdout R ²	Final Validation Loads
\$107.70	\$648.20	0.8202	12,000

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1. Objective

The objective of this project is to develop a robust, production-ready supervised machine learning regression model capable of accurately predicting freight posted rates for unseen transportation loads. Freight spot pricing operates in an inherently volatile and dynamic environment where pricing is governed by multi-dimensional interactions between haul distance, geographic origin-destination lanes, trailer equipment requirements, freight weight, temporal seasonality, and macroeconomic spot market indices.

The labeled development dataset (train_test.csv) serves as the foundation to audit data quality, conduct thorough exploratory data analysis (EDA), engineer high-signal domain features, formulate a leakage-free chronological validation strategy, evaluate diverse model architectures, and tune the top-performing regression system.

The final trained model is applied to generate rate predictions for all 12,000 unseen loads in validation.csv, strictly formatted per validation_predictions_template.csv (load_id, predicted_rate). In addition, a fixed 31-day December 2025 forecasting scenario is evaluated and validated against the supplied automated evaluation scorer.

End-to-End Machine Learning Pipeline: Data Ingestion → Data Quality Audit → Exploratory Data Analysis → Feature Engineering (27 Predictors) → Chronological Validation Setup → Baseline & Model Benchmarking → Hyperparameter Optimization → Full-Dataset Refitting → 12,000 Validation Predictions → December Scenario Forecast & Validation.

2. Dataset Architecture & Inventory

The project relies on four core data files spanning historical observations and unseen evaluation sets:

Dataset / File	Rows	Columns	Purpose / Description
train_test.csv	48,000	14	Labeled historical development data (Jan 1, 2025 – Oct 31, 2025)
validation.csv	12,000	13	Unseen out-of-time evaluation loads (Nov 1, 2025 – Dec 31, 2025)
validation_predictions_template.csv	12,000	2	Standardized submission template containing load_id and predicted_rate
december_chart_inputs.csv	31	7	Fixed load configuration across all 31 days of December 2025

2.1 Feature Definitions & Target Characteristics

The historical development dataset encompasses 14 raw features describing geographic, operational, market, and temporal characteristics:

Feature	Description	Data Type / Scale
load_id	Unique identifier for each freight load transaction	Categorical (ID)
pickup	Origin city / freight pickup terminal (64 unique locations)	Categorical
delivery	Destination city / delivery terminal (64 unique locations)	Categorical
pickup_lat	Geographic latitude coordinate of pickup facility	Numerical (Degrees)
pickup_lon	Geographic longitude coordinate of pickup facility	Numerical (Degrees)
delivery_lat	Geographic latitude coordinate of delivery facility	Numerical (Degrees)
delivery_lon	Geographic longitude coordinate of delivery facility	Numerical (Degrees)
distance	Total haul distance between pickup and delivery locations	Numerical (Miles)
equipment	Equipment type required (Dry Van, Reefer, Flatbed)	Categorical (3 classes)
weight	Reported total weight of the freight shipment	Numerical (Pounds, lb)
date	Date freight load was posted / scheduled	Datetime (YYYY-MM-DD)
market_index	Macroeconomic freight rate spot market condition indicator	Numerical (Index)
quote_signal	Proprietary quoting signal / market demand indicator	Numerical (Continuous)
posted_rate	Supervised target: actual dollar rate posted for the load	Target (\$ USD)

Target Distribution Summary: The target variable posted_rate exhibits a sample mean of \$2,373.98, median of \$2,030.76, standard deviation of \$1,486.49, minimum of \$57.22, and maximum of \$25,533.00 across the 48,000 development observations. The substantial spread and long upper tail highlight significant skewness in raw freight pricing.

3. Data Quality & Integrity Audit

Before model construction, a systematic data-quality audit was performed to evaluate missing values, schema consistency, duplicate IDs, numerical ranges, and potential outliers across the development and validation datasets.

The main missing values were found in weight and market_index, while pickup, delivery, and equipment contained no missing values. A notable anomaly was also identified in weight, where negative values were present. Since there was insufficient domain information to determine their validity, these observations were documented rather than arbitrarily removed.

The load_id field was treated as a unique identifier and excluded from model training. Overall, the audit ensured that the datasets were structurally consistent and suitable for preprocessing, feature engineering, and model development.

3.1 Missing Value Analysis

Missing values were isolated strictly to weight and market_index across both datasets. All geographic, equipment, route, and date fields had 100% data completeness:

Feature	Training Missing (out of 48,000)	Validation Missing (out of 12,000)
weight	300 (0.62%)	165 (1.38%)
market_index	374 (0.78%)	249 (2.08%)
pickup / delivery	0 (0.00%)	0 (0.00%)
coordinates (lat/lon)	0 (0.00%)	0 (0.00%)
equipment	0 (0.00%)	0 (0.00%)

Handling Strategy: For linear and standard tree baselines (Ridge, Random Forest), numerical missing values were imputed via median values computed strictly on the training partition to prevent data leakage. For the final CatBoost architecture, the model's native missing value handling mechanism was leveraged, allowing tree splits to assign missing indicators optimally.

3.2 Weight Anomaly Investigation

The weight attribute exhibited values spanning from approximately -47,500 lb to +47,500 lb. Negative freight weight is physically counter-intuitive from a commercial logistics standpoint. However, in the absence of explicit domain metadata indicating whether negative values represented tare weights, signed net differential measurements, or data recording conventions, arbitrary row deletion or ad-hoc absolute value flipping was avoided. Fabricating replacements would risk introducing artificial structural bias into unseen validation evaluations. The anomaly was preserved and documented, enabling nonlinear tree splits to naturally isolate these records.

3.3 Cardinality & Datetime Standardization

The date column was parsed into ISO datetime timestamps. Cardinality inspections confirmed exactly 64 unique pickup terminals and 64 unique delivery terminals across both training and validation sets, confirming zero cold-start geographic unseen labels in validation and validating route-level combination features.

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to understand the underlying structure of the freight-rate data and identify the variables most strongly associated with posted rates. The analysis examined statistical relationships,

pricing patterns, and potential nonlinear dependencies across physical load characteristics, geographic information, equipment types, and calendar time.

Attention was given to the distribution of posted rate, its relationship with transportation distance and weight, differences in pricing across equipment categories, and variations across pickup and delivery locations. The analysis also examined how freight rates changed over time to identify potential seasonal or calendar-based patterns.

One of the strongest findings was the relationship between **distance and posted rate**, with a Pearson correlation of approximately **0.9085**, indicating that longer transportation distances are generally associated with higher freight rates. However, the relationship was not perfectly linear, suggesting that additional factors such as equipment type, route, weight, market conditions, and date also influence pricing.

The target variable was also found to be **right-skewed**, with a relatively small number of high-value loads creating a long upper tail. This observation later motivated the use of a logarithmic transformation of the target during CatBoost modeling.

Overall, the EDA provided the basis for subsequent feature engineering and model selection by identifying the most relevant predictive signals and highlighting nonlinearities and distribution characteristics that the machine learning models needed to capture.

4.1 Distribution of Posted Rate

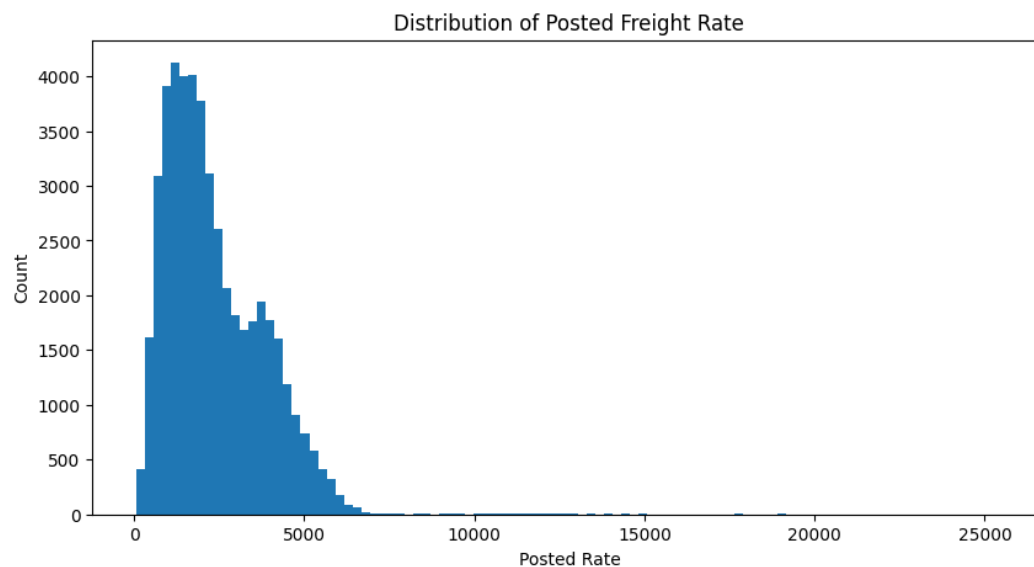


Figure 1: Empirical Distribution of Posted Freight Rate (\$ USD)

As shown in Figure 1, `posted_rate` is strongly right-skewed with a sharp concentration between \$500 and \$3,500, followed by a long tail extending past \$20,000. Fitting regression models directly on raw dollar values using mean squared error losses causes optimization gradients to be dominated by extreme tail transactions, resulting in systemic under-prediction across the high-density middle-market loads. This provided direct empirical justification for adopting a `log1p` target transformation.

4.2 Haul Distance vs. Posted Rate

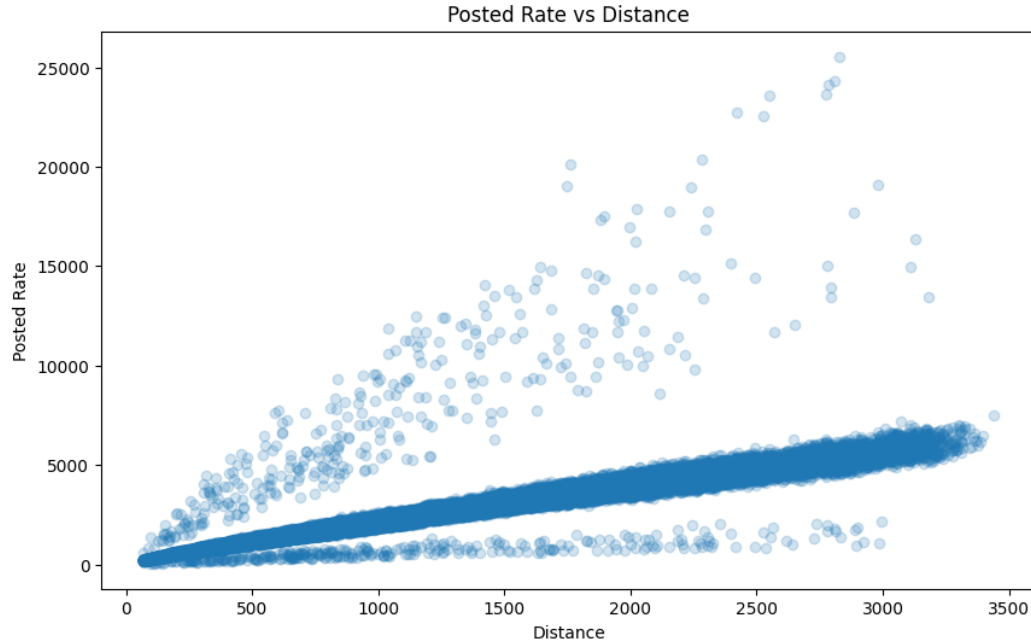


Figure 2: Scatter Plot of Posted Rate (\$) vs. Haul Distance (Miles)

Haul distance emerged as the primary numerical pricing driver, yielding a Pearson correlation coefficient of 0.9085 with `posted_rate`. Figure 2 illustrates that while rates scale monotonically with distance, the variance and upper-bound dispersion increase noticeably on long-haul routes (>1,500 miles). This heteroscedastic spread proves that distance alone is insufficient; route-specific pricing premiums, equipment premiums, and fuel surcharges exert expanding influence on long-haul routes.

4.3 Shipment Weight vs. Posted Rate

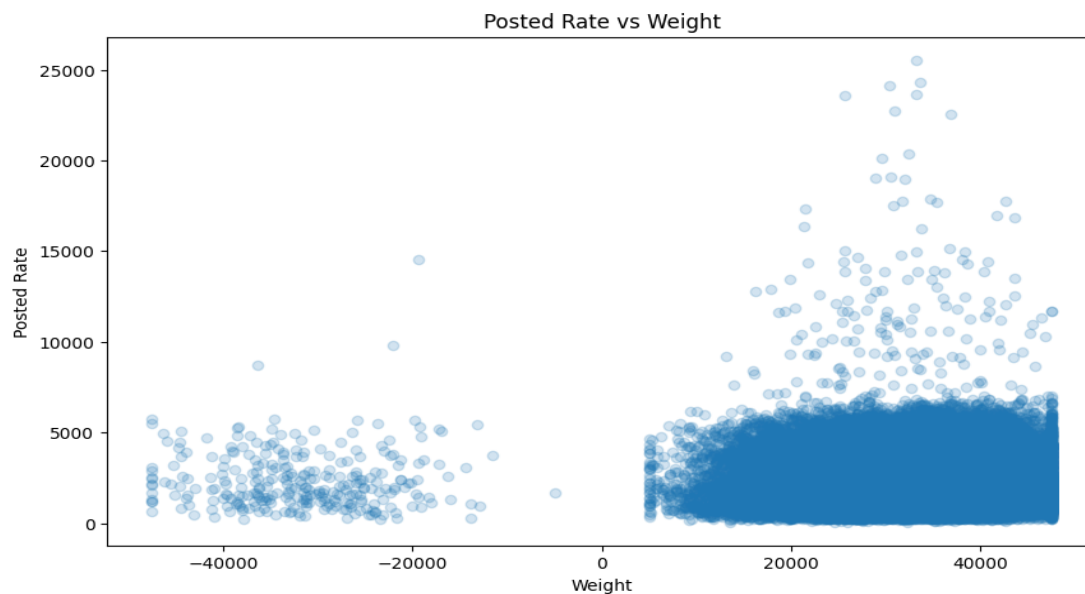


Figure 3: Scatter Plot of Posted Rate (\$) vs. Shipment Weight (lb)

Figure 3 highlights the relationship between load weight and freight pricing. In contrast to distance, weight exhibits a much weaker direct correlation with posted rates. Identically weighted shipments display wide rate variation because haul distance and route lanes dictate the baseline quote. Noticeably, the negative weight cluster mirrors the positive weight distribution across rate ranges, confirming the necessity of interaction terms (e.g., weight-to-distance ratio).

4.4 Equipment Category Impact

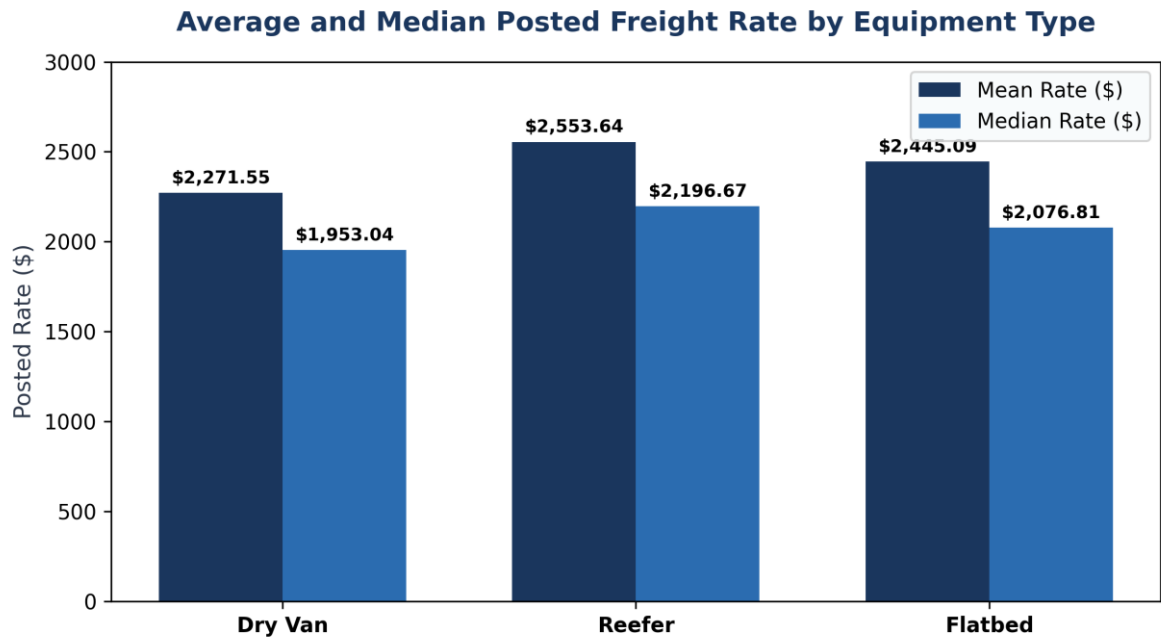


Figure 4: Average and Median Posted Freight Rate by Equipment Type (\$ USD)

Equipment type significantly influences baseline pricing due to specialized handling requirements, capital expenditure, and operating fuel costs (e.g., continuous reefer fuel consumption):

Equipment Type	Total Load Count	Mean Rate (\$)	Median Rate (\$)
Dry Van	27,202 (56.7%)	\$2,271.55	\$1,953.04
Reefer (Refrigerated)	12,045 (25.1%)	\$2,553.64	\$2,196.67
Flatbed	8,753 (18.2%)	\$2,445.09	\$2,076.81

Refrigerated (Reefer) loads command the highest premium with a median rate of \$2,196.67, reflecting refrigeration operating costs, while standard Dry Van loads maintain the lowest median of \$1,953.04.

4.5 Geographic Origin & Destination Dynamics

Analysis of the 64 pickup and 64 delivery facilities reveals stark regional pricing differentials. High-cost freight hubs, particularly major West Coast freight centers, exhibit median rates over \$4,000, whereas Midwest and Southern regional hubs command significantly lower median rates:

Top Pickup Hubs	Pickup Median (\$)	Top Delivery Hubs	Delivery Median (\$)
San Francisco	\$4,301.81	San Francisco	\$4,356.60
Fresno	\$4,160.28	Fresno	\$4,140.17
Los Angeles	\$3,996.61	Los Angeles	\$4,046.09
Phoenix	\$3,935.62	Phoenix	\$3,921.98
Reno	\$3,836.93	Reno	\$3,783.88

4.6 Temporal Pricing Dynamics

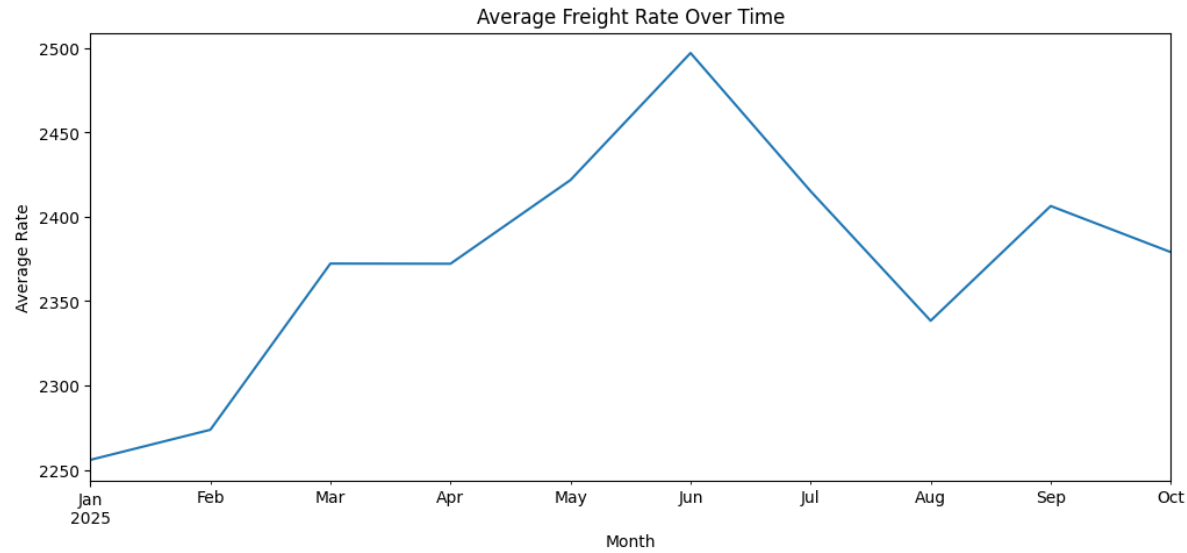


Figure 5: Average Freight Rate Trend Across Historical Months (Jan–Oct 2025)

Figure 5 tracks the mean monthly freight rate across the development timeline. Rates demonstrate pronounced seasonality, rising steadily from \$2,256 in January to a peak near \$2,500 in June, before experiencing late-summer contraction and an autumn resurgence. This empirical trend confirms that time is an active predictive driver, necessitating structured calendar decomposition and cyclical trigonometric encodings.

5. Feature Engineering Architecture

To enrich the raw tabular signals and provide the gradient-boosted model with meaningful representations of the underlying freight-pricing factors, a comprehensive feature-engineering pipeline was developed. The pipeline generated **27 predictive features**, excluding the unique identifier `load_id` and the target variable `posted_rate`.

The engineered features were designed to capture several dimensions of the freight-pricing problem, including **geographic information, physical load characteristics, route-level patterns, temporal effects, nonlinear relationships, and feature interactions**. Rather than relying only on the original columns, transformed representations were introduced to help the model capture relationships that may not be easily represented by the raw variables alone.

The date variable was decomposed into multiple calendar features such as year, month, day, day of week, day of year, and week of year. Cyclical sine and cosine transformations were additionally created to represent recurring calendar patterns. Distance and weight were transformed using logarithmic representations, while squared distance was introduced to capture nonlinear distance effects.

A combined route feature was also created from the pickup and delivery locations, allowing the model to learn route-specific pricing patterns. Furthermore, `weight_distance_ratio` was introduced to represent the interaction between the physical size of a load and the distance it needs to travel.

Feature Category	Feature Count	Engineered Variable Names & Formulations
Geographic Identifiers	2	<code>pickup</code> , <code>delivery</code> (categorical location IDs)
Spatial Coordinates	4	<code>pickup_lat</code> , <code>pickup_lon</code> , <code>delivery_lat</code> , <code>delivery_lon</code>
Physical Shipment	2	<code>distance</code> , <code>weight</code>
Equipment Specification	1	<code>equipment</code> (Dry Van, Reefer, Flatbed)
Macro Market Signals	2	<code>market_index</code> , <code>quote_signal</code>
Temporal Calendar	6	<code>year</code> , <code>month</code> , <code>day</code> , <code>dayofweek</code> , <code>dayofyear</code> , <code>weekofyear</code>
Cyclical Temporal	4	<code>sin_doy</code> , <code>cos_doy</code> (annual), <code>sin_dow</code> , <code>cos_dow</code> (weekly)
Nonlinear Transformations & Interactions	6	<code>distance_log</code> , <code>distance_sq</code> , <code>weight_log</code> , <code>weight_distance_ratio</code> , <code>route</code> (<code>pickup</code> → <code>delivery</code>)

This feature-engineering architecture allowed the final model to use both the original operational information and additional representations of important relationships. In particular, the multiple distance representations were valuable because EDA had identified distance as the strongest individual numerical signal. The engineered features therefore provided the model with a richer representation of freight demand and pricing behavior while retaining the original information needed for prediction.

5.1 Cyclical & Interaction Formulations

- **Cyclical Calendar Trigonometry:** Standard linear day-of-year integers fail to reflect that December 31 and January 1 are adjacent. Cyclical encoding via $\sin(2\pi \cdot \text{doy} / 365.25)$ and $\cos(2\pi \cdot \text{doy} / 365.25)$ maps continuous seasonal progression onto a seamless circle.
- **Nonlinear Distance Representations:** Engineered logarithmic distance ($\log(1 + \text{distance})$) and quadratic distance (distance^2) enable tree nodes to partition short-haul versus cross-country hauls efficiently.
- **Physical Load Density Interaction:** The `weight_distance_ratio` ($\text{weight} / (\text{distance} + 1)$) captures ton-mile freight density dynamics, providing signal on load transport intensity.
- **Composite Route Categorical:** Combining pickup and delivery into an explicit route string (e.g., `'Lexington_to_Fort Wayne'`) allows CatBoost's target encoding to learn lane-specific rate equilibria directly.

5.2 Feature Importance Analysis

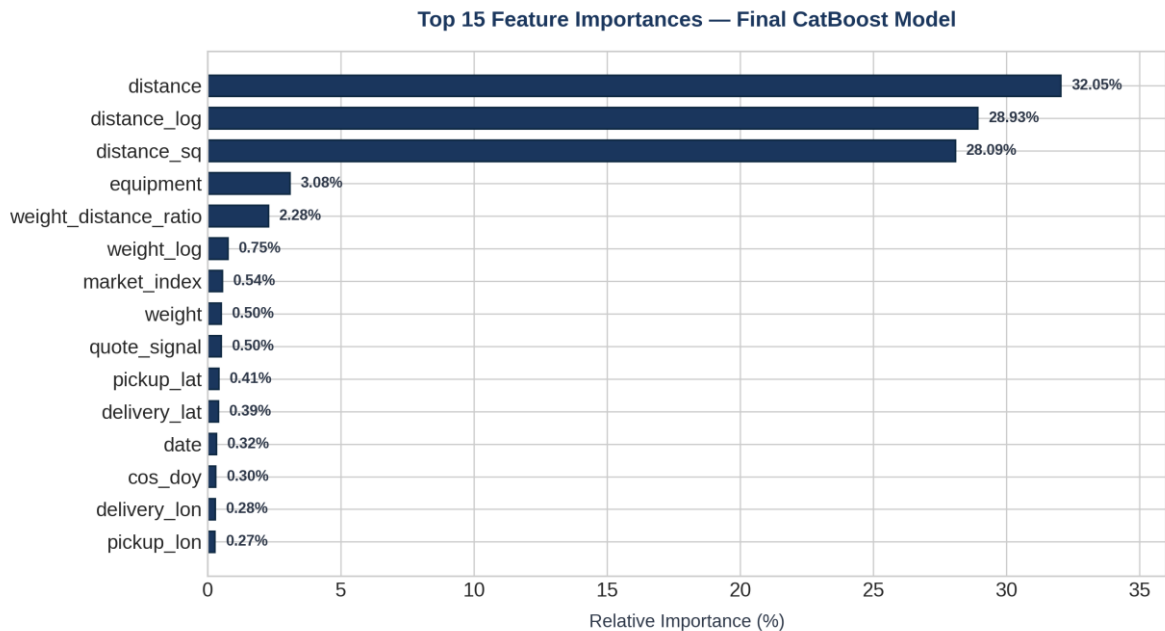


Figure 6: Top 15 Feature Importances in Final Tuned CatBoost Model (%)

As quantified in Figure 6, distance-derived features overwhelmingly dominate the model's decision architecture. Raw distance (32.05%), log distance (28.93%), and quadratic distance (28.09%) together constitute 89.07% of the total model split importance, confirming the EDA finding. Equipment type (3.08%) and the weight-distance interaction ratio (2.28%) provide secondary predictive power, followed by weight transforms, market indicators, and geographic coordinates.

6. Validation Strategy & Metric Selection

A fundamental requirement for applying machine learning models to freight logistics is **temporal generalization**: a model trained on historical market data should be able to predict future load transactions without gaining access to information from the future. To reflect this real-world setting, a **chronological validation strategy** was adopted instead of a conventional random train-test split.

The labelled development dataset covers freight transactions from **January 1, 2025, through October 31, 2025**. The data was divided on **October 1, 2025**, resulting in an earlier training period and a later holdout period. This approach ensures that the model learns only from observations that would have been available before the validation period.

6.1 Leakage-Free Chronological Holdout

Random K-Fold cross-validation was strictly rejected because shuffling freight transactions across time allows future market conditions to bleed into training folds, yielding unrealistically optimistic error metrics. The 48,000-row development dataset was split chronologically at October 1, 2025:

Partition	Date Range	Row Count	Operational Role
Training Set	January 1, 2025 – September 30, 2025	43,147 rows	Model fitting and feature transformation learning
Validation Holdout	October 1, 2025 – October 31, 2025	4,853 rows	Out-of-time evaluation & model selection

6.2 Evaluation Metrics

- Mean Absolute Error (MAE) [Primary Metric]:** $MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$
Selected as the **primary model-selection metric** because it directly represents the average absolute prediction error in dollars. A lower MAE indicates that the predicted freight rates are, on average, closer to the actual posted rates.
- Root Mean Squared Error (RMSE):** $RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$
RMSE was used as a secondary metric. Unlike MAE, RMSE gives greater weight to larger prediction errors, making it useful for identifying models that occasionally produce substantial deviations from the actual freight rate.
- Coefficient of Determination (R²):** $R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$
This was used to measure how much of the variation in posted freight rates is explained by the model. Higher R² values indicate stronger overall explanatory performance.

7. Model Exploration & Benchmark Comparison

Model development followed a progressive benchmarking approach, beginning with a simple median-rate baseline and advancing to regularized linear regression, tree-based ensemble learning, and gradient-boosted trees.

The **Median Baseline** provided a reference point, while **Ridge Regression** evaluated linear relationships. **Random Forest** was then used to capture nonlinear patterns, followed by **CatBoost** to better handle categorical features and complex interactions.

Finally, CatBoost was evaluated using both the raw target and a **log-transformed target**, with the latter producing the strongest overall validation performance. This progressive comparison helped quantify the improvement gained from increasing model complexity and informed the selection of the final modeling approach.

Model Architecture	Holdout MAE (\$) ↓	Holdout RMSE (\$) ↓	Holdout R² ↑
Median Baseline Heuristic	\$1,146.79	\$1,567.97	-0.0522
Ridge Regression (L2 Linear)	\$410.82	\$782.83	0.7377
Random Forest (100 Trees)	\$188.51	\$704.23	0.7877
CatBoost Regressor (Raw Target)	\$140.00	\$649.57	0.8194

Model Architecture	Holdout MAE (\$) ↓	Holdout RMSE (\$) ↓	Holdout R ² ↑
CatBoost Regressor (Log-Transformed Target)	\$108.68	\$646.50	0.8211

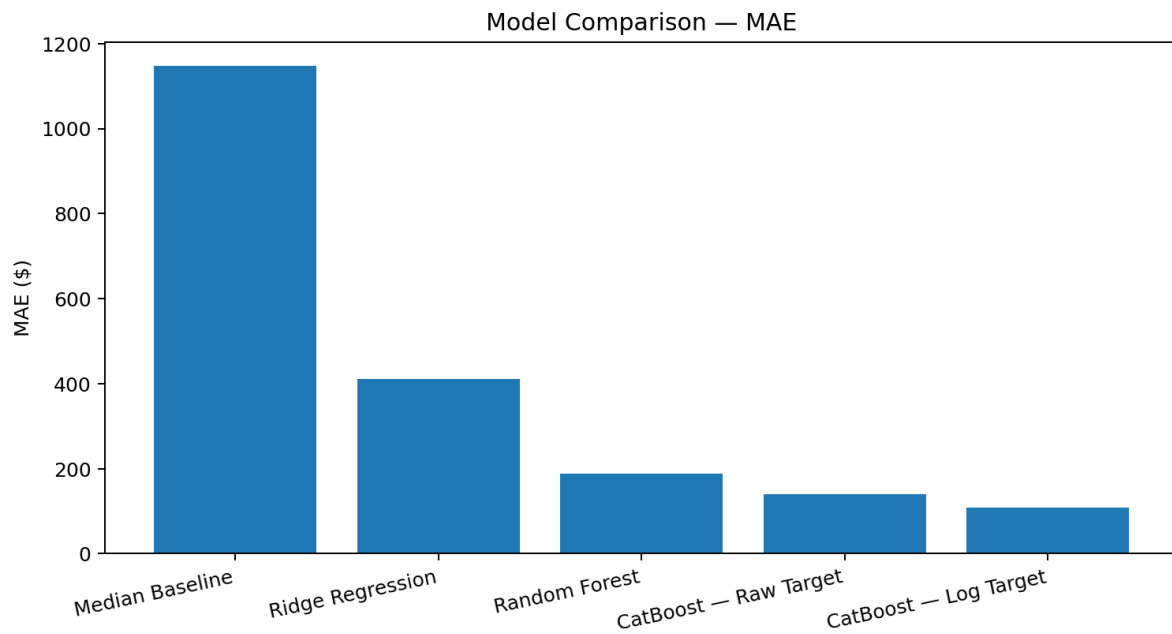


Figure 7: Model Comparison — Mean Absolute Error (MAE in \$ USD, Lower is Better)

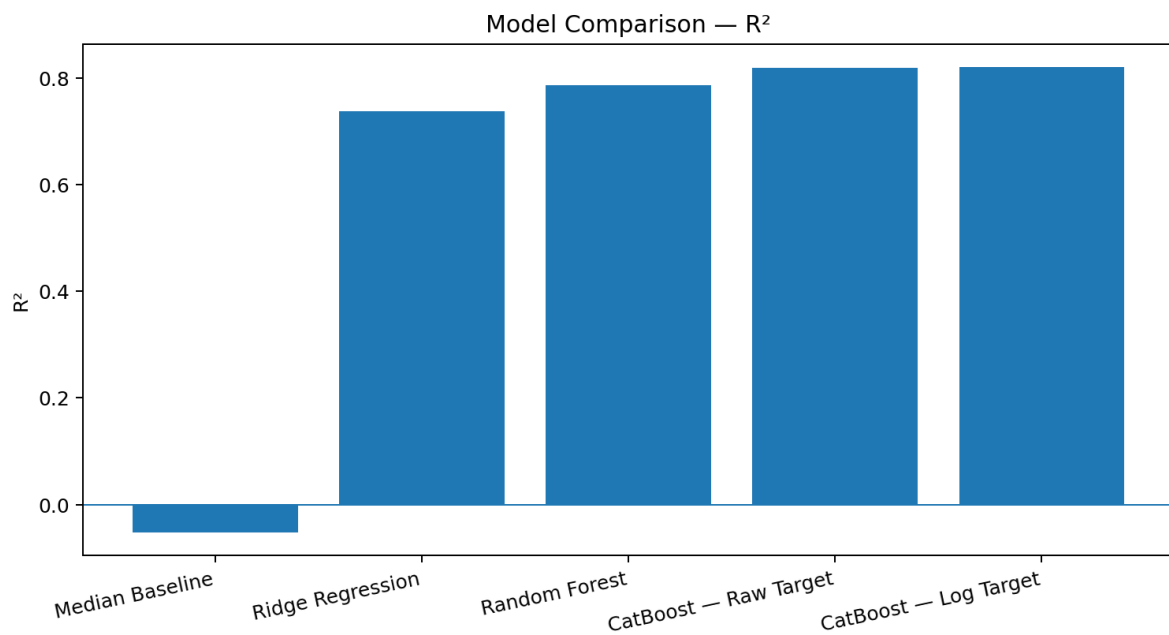


Figure 8: Model Comparison — Coefficient of Determination (R², Higher is Better)

7.1 Progression Analysis & Key Takeaways

- 1. Linear Limitations: Ridge regression established a substantial improvement over the median baseline (MAE dropping from \$1,146.79 to \$410.82), confirming the strength of distance signals. However, its inability to model nonlinear route interactions left significant residual error.
- 2. Nonlinear Tree Ensembles: Random Forest slashed MAE to \$188.51, confirming that localized route, equipment, and weight thresholds require decision-tree logic.
- 3. Gradient Boosting & Native Categoricals: CatBoost with raw targets reduced MAE further to \$140.00 by utilizing ordered boosting and efficient target-based categorical encoding on route, pickup, delivery, and equipment.
- 4. Target Log-Transformation Breakthrough: Transforming the target via log1p(posted_rate) achieved the single largest structural gain, dropping MAE from \$140.00 to \$108.68. Compressing the long right tail stabilized gradient updates and prevented rare multi-thousand-dollar loads from distorting the model's loss landscape.

8. Hyperparameter Optimization

Building upon the log-transformed CatBoost architecture, targeted hyperparameter searches were performed on the same chronological holdout dataset. The optimization focused on key parameters controlling model complexity and learning behaviour: **boosting iterations, tree depth, learning rate, and L2 leaf regularization**. Multiple parameter combinations were evaluated using **MAE, RMSE, and R²**. MAE was used as the primary selection criterion, consistent with the validation strategy established earlier.

The best-performing configuration used **300 iterations, depth 6, a learning rate of 0.05, and L2 leaf regularization of 5**, achieving an MAE of **107.70** on the chronological holdout. Although some deeper configurations produced slightly better RMSE and R² values, the selected configuration achieved the lowest MAE and was therefore chosen as the final model.

Iterations	Depth	Learning Rate	L2 Leaf Reg	Holdout MAE (\$)	Holdout RMSE (\$)	Holdout R ²
300	6	0.05	5	\$107.70	\$648.20	0.8202
500	8	0.05	5	\$108.68	\$646.50	0.8211
700	8	0.03	8	\$108.53	\$646.33	0.8212
600	10	0.04	8	\$108.35	\$645.98	0.8214

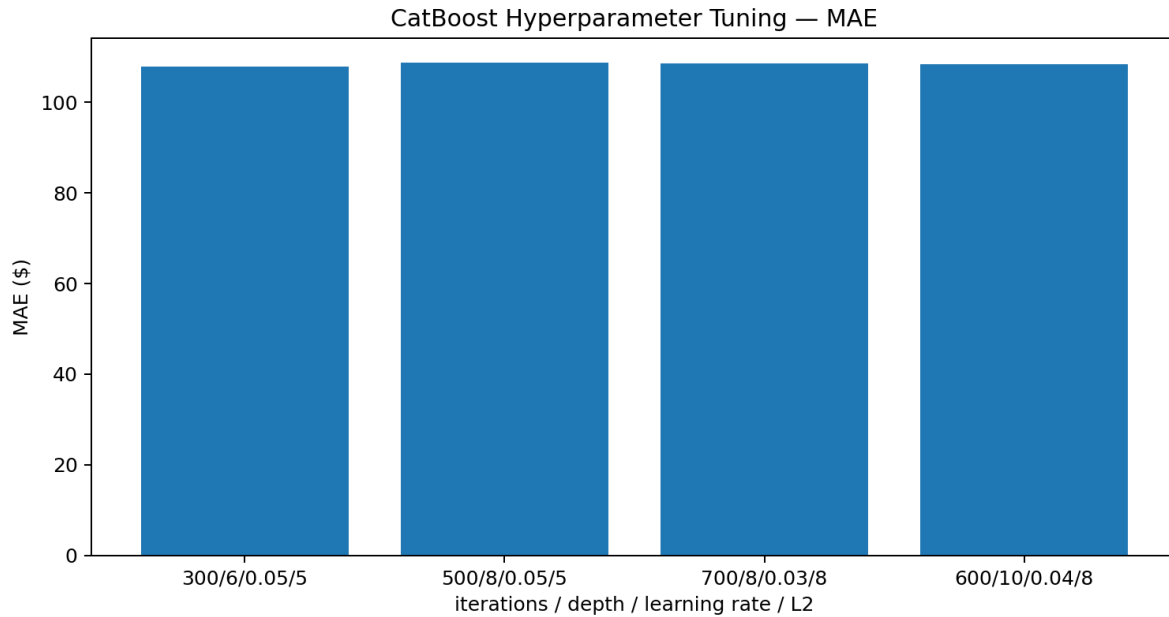


Figure 9: CatBoost Hyperparameter Tuning — Holdout MAE Comparison (\$ USD)

8.1 Final Configuration Selection

The configuration utilizing 300 iterations, tree depth 6, learning rate 0.05, and L2 regularization 5 achieved the lowest validation MAE of \$107.70. While deeper architectures (depth 10, 600 trees) achieved marginal improvements in RMSE (\$645.98) and R^2 (0.8214), MAE was pre-defined as the decisive primary business metric. Furthermore, the shallower depth also reduces model complexity and is expected to provide lower inference cost than the deeper alternatives.

9. Final Production Model Specification

The final configuration selected for inference was the **CatBoostRegressor with a log-transformed target**. The model was selected based on its performance on the chronological holdout, with **MAE used as the primary selection metric**.

Model Specification	Production Value / Setting
Algorithm	CatBoostRegressor (Yandex Open Source)
Target Transformation	log1p(posted_rate) forward; expm1(prediction) inverse
Boosting Iterations	300 trees
Maximum Tree Depth	6
Learning Rate (Shrinkage)	0.05
L2 Leaf Regularization	5.0

Model Specification	Production Value / Setting
Loss Objective	RMSE on log-space target
Categorical Features	pickup, delivery, equipment, route

For training, the target variable was transformed using `log1p()` to reduce the influence of the highly right-skewed freight-rate distribution. After prediction, the values were converted back to the original rate scale using the inverse `expm1()` transformation.

Following model selection, the final configuration was retrained using the complete **48,000-row labeled development dataset** before generating predictions for the 12,000 unseen validation loads.

9.1 Full-Dataset Retraining & Validation Predictions

Following hyperparameter selection, the final CatBoost configuration was refitted on the complete 48,000 labeled development observations (January–October 2025). Retraining on 100% of labeled data maximizes sample exposure and route frequency distributions before predicting unseen loads.

The refitted model was executed across all 12,000 unseen loads in `validation.csv`. The generated predictions satisfy all structural checks:

- **Record Count:** Exactly 12,000 predictions generated matching validation load IDs.
- **Prediction Range:** Minimum predicted rate is \$205.57; maximum predicted rate is \$6,786.45.
- **Integrity:** Zero nulls, zero NaN values, and zero non-positive values.

10. Validation Diagnostics & Error Analysis

Evaluating the tuned model on the October chronological holdout established the definitive development performance benchmark. The selected CatBoost configuration achieved strong predictive performance across all three-evaluation metrics, with **MAE used as the primary model-selection criterion**.

Validation Metric	Holdout Benchmark Score
Mean Absolute Error (MAE)	\$107.7028 per load
Root Mean Squared Error (RMSE)	\$648.1977
Coefficient of Determination (R²)	0.8202 (82.02% variance explained)

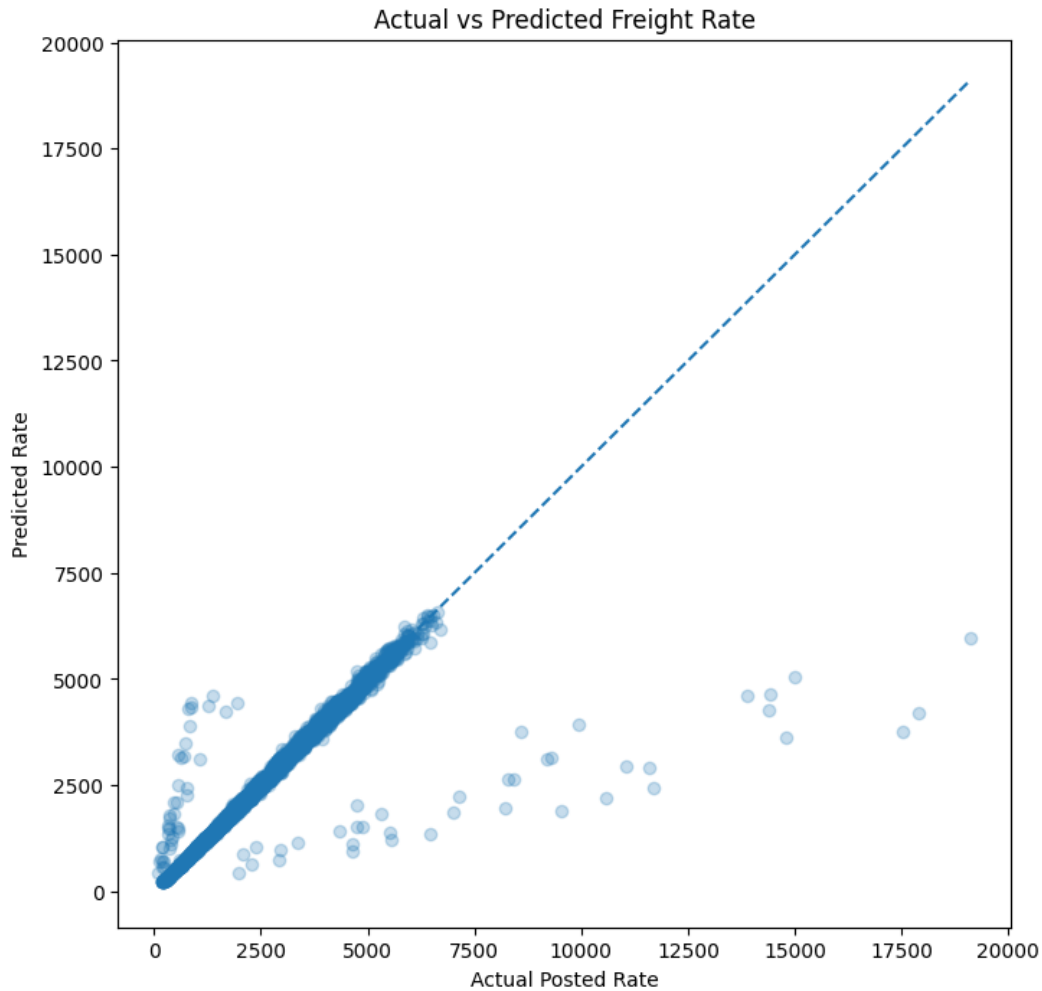


Figure 10: Actual vs. Predicted Freight Rates on Chronological Holdout (\$ USD)

The MAE of approximately **\$107.70** indicates that the model's predictions differ from the actual posted freight rates by an average absolute amount of around \$108 on the October holdout.

The RMSE of **\$648.20** is considerably higher than the MAE, indicating that a smaller number of observations produce relatively larger prediction errors. This is consistent with the right-skewed target distribution and the presence of high-value freight loads.

The **R² score of 0.8202** indicates that the model explains approximately 82% of the variance in posted freight rates within the chronological holdout.

10.1 Diagnostic Interpretation

Figure 10 displays actual versus predicted freight rates against the 45-degree identity reference line. Predictions track the ideal trajectory across the dense core between \$500 and \$6,500. Deviations expand primarily in high-rate outliers above \$7,500, which accounts for the spread between MAE (\$107.70) and RMSE (\$648.20). Importantly, development holdout metrics represent rigorous out-of-time benchmarks and are distinct from final hidden test scores calculated post-submission by Spotter.

11. Fixed December 2025 Scenario Reconstruction

The assessment defines a fixed operational stress-test across all 31 days of December 2025 (december_chart_inputs.csv) holding haul parameters constant:

Fixed Operational Parameter	Scenario Value
Pickup City (Origin)	Lexington
Delivery City (Destination)	Fort Wayne
Haul Distance	360 miles
Equipment Specification	Dry Van
Shipment Weight	32,000 lb
Forecast Date Range	December 1, 2025 – December 31, 2025 (31 days)

Because the December input file does not contain every feature required by the trained model, the missing geographic and market-related inputs were reconstructed using information from the development dataset. Historical location-level information was used for the pickup and delivery coordinates, while recent development-period values were used for market_index and quote_signal.

After reconstructing these inputs, the same feature-engineering pipeline used during model training was applied to the December data. This ensured consistency between the training and inference feature structures, including the calendar, cyclical, distance, weight, interaction, and route features.

The resulting dataset contained **31 daily observations with all required model features available**. The final CatBoost model was then used to generate a predicted freight rate for each day of December 2025.

This scenario provides a controlled way to examine how the model's predicted rate changes over time while keeping the underlying load characteristics—route, distance, equipment, and weight—constant.

11.1 Feature Imputation & Generation Pipeline

The raw December scenario file omitted geographic coordinates and macro market indices. To ensure feature consistency without lookahead bias, missing variables were reconstructed using historical development medians:

- **Terminal Coordinates:** Lexington pickup coordinates were mapped to historical median (lat: 36.99152, lon: -84.99876); Fort Wayne delivery coordinates were mapped to (lat: 41.31561, lon: -85.36206).
- **Market Signals:** market_index (0.95818) and quote_signal (1.98584) were imputed from recent development observations (October 2025 onward).
- **Feature Alignment:** All 27 engineered predictors were generated; zero nulls remained in the December matrix.

12. December Predictions & Candidate Chart

The final CatBoost model generated daily predicted rates for all 31 days of December 2025:

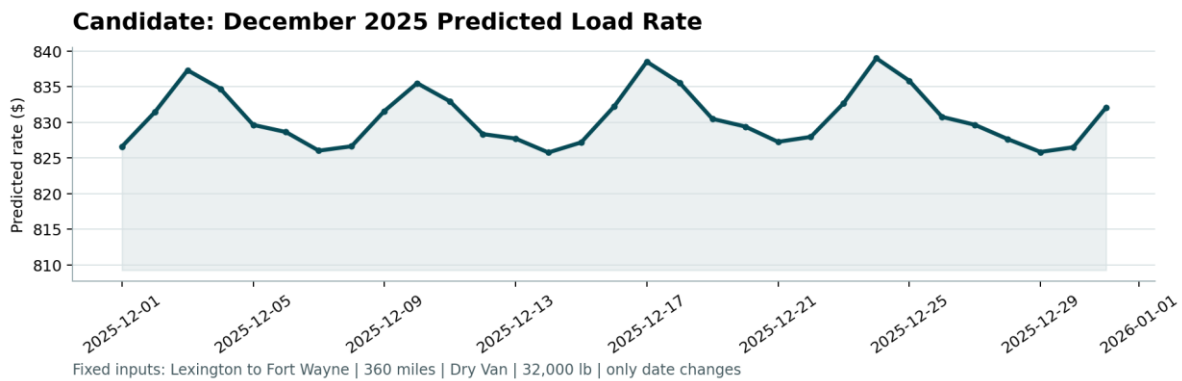


Figure 11: Candidate December 2025 Predicted Load Rate Trend (\$ USD)

12.1 Daily Rate Schedule

Daily predicted rates remain tightly bounded between \$825.75 and \$838.97, reflecting fixed physical load attributes while responding to day-of-week and month-end freight market cycles:

Date	Rate (\$)	Date	Rate (\$)
2025-12-01	\$826.58	2025-12-17	\$838.45
2025-12-02	\$831.42	2025-12-18	\$835.50
2025-12-03	\$837.21	2025-12-19	\$830.52
2025-12-04	\$834.65	2025-12-20	\$829.40
2025-12-05	\$829.62	2025-12-21	\$827.25
2025-12-06	\$828.60	2025-12-22	\$827.95
2025-12-07	\$826.05	2025-12-23	\$832.65
2025-12-08	\$826.65	2025-12-24	\$838.97
2025-12-09	\$831.54	2025-12-25	\$835.80
2025-12-10	\$835.48	2025-12-26	\$830.78
2025-12-11	\$833.02	2025-12-27	\$829.60
2025-12-12	\$828.40	2025-12-28	\$827.60
2025-12-13	\$827.68	2025-12-29	\$825.80

Date	Rate (\$)	Date	Rate (\$)
2025-12-14	\$825.75	2025-12-30	\$826.45
2025-12-15	\$827.18	2025-12-31	\$832.05
2025-12-16	\$832.25		

Automated Scorer Verification Log: Execution of score.py confirmed 100% compliance across all evaluation checks:

- Validated 12,000 final predictions (correct headers, valid load IDs, strictly positive rates).
- Validated 31 fixed December predictions (correct chronology, fixed input invariants maintained).
- Successfully generated verification visualization: scorer_results/candidate_december.png.

13. Conclusion & Assessment Summary

This assessment implemented a rigorous, production-grade machine learning workflow for freight spot rate prediction:

- **Systematic Exploratory Findings:** Identified haul distance as the primary linear driver ($r = 0.9085$), recognized significant equipment premiums (Reefer loads carrying a \$243.63 median premium over Dry Van), and proved the presence of high right-skewness in spot freight pricing.
- **Domain-Specific Feature Engineering:** Constructed 27 predictive features combining geographic origin-destination pairs, latitude/longitude coordinates, cyclical calendar harmonics, nonlinear distance transforms, and load-density interaction terms.
- **Principled Chronological Validation:** Implemented an out-of-time evaluation split (January–September training vs. October holdout) to prevent future lookahead leakage and evaluate true forward temporal generalization.
- **Benchmark-Driven Model Selection:** Progressed from a naive baseline (\$1,146.79 MAE) through Ridge (\$410.82 MAE), Random Forest (\$188.51 MAE), raw CatBoost (\$140.00 MAE), and log-transformed CatBoost (\$108.68 MAE).
- **Hyperparameter Optimization:** Tuned CatBoost (300 iterations, depth 6, learning rate 0.05, L2 regularization 5) to achieve the top validation MAE of \$107.70 per load on the chronological holdout.
- **Submission Compliance:** Successfully generated all 12,000 unseen validation predictions and 31 December scenario forecasts, with automated verification confirming zero data errors, complete schema adherence, and valid positive predictions.

Candidate Submission Information

Candidate Name:	Sayyad Emad Quadri
GitHub Repository:	https://github.com/equadri17/freight-rate-prediction
Walkthrough Loom URL:	[Insert Loom URL before final submission]

14. Limitations & Future Improvements

14.1 Limitations

The model demonstrates strong performance on the chronological development holdout, but the October holdout is still a single temporal evaluation window. Therefore, performance may vary under substantially different future market conditions.

The dataset also contains anomalous negative weight values. These observations were intentionally preserved because sufficient domain information was unavailable to justify correcting or removing them.

The December scenario required reconstruction of missing market signals using recent historical values. Consequently, December predictions should be interpreted as a controlled scenario forecast rather than a fully independent market forecast.

Finally, the hidden evaluation set does not provide ground-truth posted rates during model development, so the true generalization performance on the final evaluation set cannot be established until scoring is performed by the assessment evaluator.

14.2 Future Improvements

Future work could investigate rolling-origin cross-validation, explicit route-level historical statistics, additional market and fuel-price indicators, uncertainty intervals, model assembling, and systematic feature-ablation experiments.